Answer **all** the questions in the spaces provided.

1 **(a)** **(i)** State Newton's Second Law of Motion as applied to a *system of bodies*.

.....

.....

..... [2]

(ii) Explain the implication of Newton's Second Law for a *system of bodies* isolated from all external forces.

.....

.....

..... [2]

(b) Two blocks A and B are held together with a light spring in between them as shown in Fig. 1.1. The spring has a force constant 80 N m^{-1} and is compressed by 0.060 m.

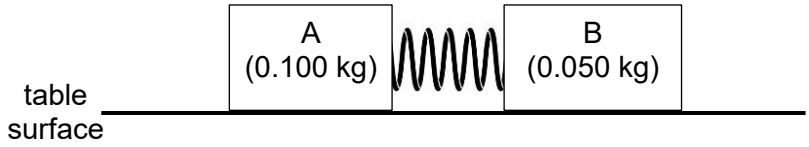

Fig. 1.1

The masses of blocks A and B are 0.100 kg and 0.050 kg respectively. Upon release, the two blocks move off in opposite directions on a smooth table surface and the spring falls off.

(i) On Fig. 1.2, sketch a graph to show the variation with time *t* of the force exerted by the spring on block B during which B is being pushed away from block A.

Fig. 1.2

[2]

(ii) Determine the final speeds of the two blocks.

speed of block A = ... m s^{-1}

speed of block B = ... m s^{-1} [4]

(c) As block B slides forward, it topples over at the edge of the table and lands on a horizontal uniform plank hinged to a wall at O. The centre of block B is at a horizontal distance of 0.15 m from the free end of the plank. The plank is supported by a rope attached to the wall as shown in Fig. 1.3.

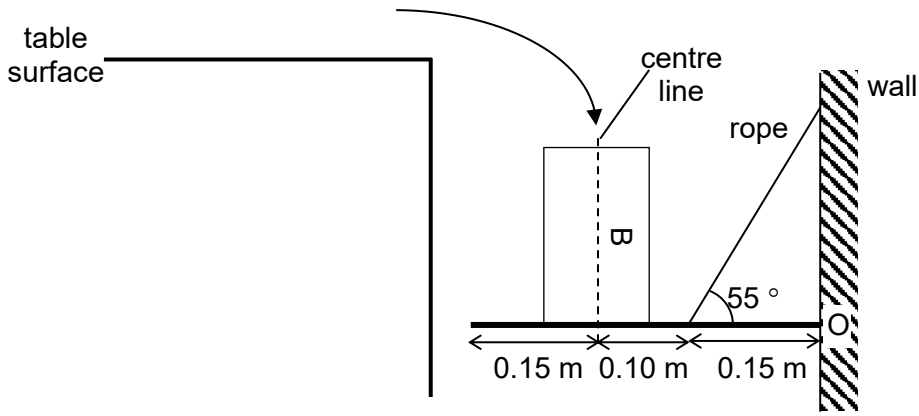

Fig. 1.3

Given that the weight of the plank is 4.0 N, calculate

(i) the tension in the rope after B is at rest on the plank,

tension = ... N [2]

(ii) the magnitude of the horizontal and vertical components of the force exerted by the hinge on the plank,

magnitude of horizontal component = ... N

magnitude of vertical component = ... N [2]

(iii) the angle θ that the force by the hinge on the plank makes with the horizontal. On Fig. 1.3, draw and label the force *F* by the hinge and indicate the angle θ.

θ = ... ° [1]